

Yang Shi

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[LinkedIn](#) | [Portfolio](#) | [GitHub](#)

SUMMARY

Gameplay and software developer with 3 years of professional software engineering experience and graduate-level game development training at SMU Guildhall. Strong C++ background in custom engine systems, DirectX 11 rendering, gameplay features, networking, tools, and Unreal Engine 5 development. Experienced building production web applications with React, Next.js, Node.js, REST APIs, databases, Docker, and CI/CD, with a focus on readable code, debugging discipline, and end-to-end feature ownership.

Game Development Experience

Hamsterballin | Unreal Engine 5, C++, Blueprints

- Built a real-time race position system that calculates live player rankings based on track progress, providing accurate placement updates throughout each race.
- Implemented a Grand Prix mode featuring multi-race series flow, cumulative point accumulation, and end-of-series result tracking.
- Designed and developed race UI elements including a live leaderboard and race-state indicators that dynamically reflect real-time gameplay data.

Custom 2D/3D Game Engine | C++, DirectX 11

- Built a custom 2D/3D game engine from scratch in C++ with DirectX 11, featuring a complete rendering pipeline and modular engine architecture.
- Implemented a modular rendering pipeline with HLSL shaders, constant buffer management, and configurable render states for flexible visual output.
- Integrated Dear ImGui as a runtime debugging UI for inspecting engine state, editing values, and improving iteration speed while developing gameplay and animation tools.

NetChess3D | C++, Custom Engine, TCP/IPv4, Winsock, DirectX 11

- Implemented chess gameplay rules in C++, including legal move validation for pawns, rooks, bishops, knights, queens, kings, captures, pawn promotion, castling support, en passant logic, and board override/debug commands.
- Built TCP networking features with Winsock using non-blocking sockets, server/client connection states, send/receive queues, null-terminated command packets, and remote DevConsole command execution.
- Created 3D board interaction systems for mouse hover, selection, legal move highlights, piece movement, promotion UI, camera cycling, and debug visualization.

Doomenstein | C++, Custom Engine, Gameplay Programming

- Built project gameplay systems inspired by classic first-person action games, including player movement, enemy interactions, map-driven gameplay, collision checks, and debug tools.
- Implemented feature code in C++ within a custom engine architecture, emphasizing readable systems, deterministic game-state updates, and rapid debugging through console commands and visualization.

Animation Viewer | C++, Assimp, glTF/FBX, DirectX 11, HLSL

- Implemented skeletal animation import and playback for glTF/FBX assets, including node hierarchy traversal, skeleton data, inverse bind matrices, animation clips, keyframe tracks, and transform sampling.
- Moved character skinning toward the GPU by passing joint indices, weights, and skin matrices to HLSL, reducing CPU-side animation cost for skinned meshes.

TECHNICAL SKILLS

- Languages: C++, JavaScript, TypeScript, ES6, Java, Python, Go, HTML5, CSS3, SASS, SQL
- Game Engine: Unreal Engine 5, Unity, custom C++ game engines
- Frontend: React.js, Next.js, Vue.js, Ember.js, Webpack, Vite
- Backend: Node.js, Express.js, GraphQL, RESTful APIs, NextAuth.js, JWT, OAuth
- Databases & ORM: MySQL, PostgreSQL, MongoDB, Firebase, Supabase, Prisma, Mongoose, Sequelize
- DevOps & Cloud: Docker, Kubernetes, AWS, Jenkins, CI/CD
- Networking: TCP/IPv4, Winsock, non-blocking sockets, client/server architecture, send/receive queues, packet parsing
- UI Libraries: Material-UI, Tailwind CSS, Bootstrap
- Tools: Git, Perforce, GitHub, Bitbucket, GitLab, Jira, Confluence, Trello, Linux/Unix, Visual Studio 2022, VS Code, IntelliJ IDEA

EXPERIENCE

- Co-Founder** | Aurora Technology and Solutions Ltd., Vancouver, BC, Canada Apr 2024 – Jul 2025
- Coordinated client projects in software and website development, delivering tailored solutions for small and medium-sized enterprises.
 - Advised clients on technology strategies to address business challenges, aligning solutions with organizational goals.
- Front End Developer** | Yun Shang Gong-Land United Technology, China Nov 2022 – Apr 2023
- Developed a tax information management interface with React.js and Taro.js, implementing responsive design that improved First Contentful Paint by 30%.
 - Contributed to e-commerce frontend and backend management systems, enhancing the end-to-end shopping experience.
 - Created a reusable React component library adopted across 3 projects, cutting frontend development time by 20%.
- Software Developer** | Standard Chartered GBS, China Sep 2021 – Sep 2022
- Developed a Positive Pay system with React.js for generating and validating electronic checks, securing transactions across designated financial institutions.
 - Rebuilt Android fingerprint authentication using Ember.js and Cordova, replacing insecure local storage with token-based session management.
 - Designed and shipped APIs for payee fund access via SMS/email notifications, serving 10,000+ daily transactions.
- Software Developer** | Jili Technology, China Mar 2021 – Aug 2021
- Reconstructed a laboratory management system with React and Ant Design, covering entrance records and chemical material tracking.
 - Designed RESTful APIs with Express.js and MySQL, implementing data segregation for multi-tenant access.
 - Built reusable UI components that reduced frontend development time by 20% across the engineering team.
- Program Analyst (Intern)** | Ontario Lottery and Gaming Corporation, Canada May 2018 – Apr 2019
- Developed intranet web pages with HTML, CSS, and Bootstrap; implemented a MySQL-backed project management system.
 - Led a barcode security initiative using C# WinForms, enhancing lab security and inventory tracking workflows.

EDUCATION

- Master of Interactive Technology in Digital Game Development** Aug 2025 – May 2026
Southern Methodist University (SMU Guildhall), Dallas, TX
- Bachelor of Science, Computer Science (Honours)** Sep 2015 – Jun 2019
Algoma University, Sault Ste. Marie, Canada